

Input and Output Screenshots for Week 4 Assignment

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DSC 350: Data Wrangling for Data Science

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June 29, 2025

1. Example 4-1:

Input 4-1

```

main.py > ...
1  #=====
2  #Title: Week 4.1 Exercise
3  #Author: Susan E. McGregor
4  #Date: 29 June 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates the use
7  #a simple example of how to read data from a .csv source_file
8  #from the Citi Bike system data
9  #=====
10
11 #Import the csv library into our program
12 import csv
13
14 #Use the open function in order to open our '202009CitibikeTripdataExample.csv'
15 #file in read mode and assign it to a variable
16 source_file = open("202009CitibikeTripdataExample.csv","r")
17
18 #Pass the 'source_file' variable that contains our Citi Bike data to the
19 #'csv' library's DictReader function, and store the result in a variable
20 #called 'citibike_reader'
21 citibike_reader = csv.DictReader(source_file)
22
23 #Use the .fieldnames attribute from the 'csv' library and
24 #Python's built in print function in order to see the individual headers
25 #from the '202009CitibikeTripdataExample.csv' file
26 print(citibike_reader.fieldnames)
27
28 #Use a for loop to iterate through the first 5 rows data
29 for i in range(0,5):
30     print(next(citibike_reader))

```

Output 4-1

```

['tripduration', 'starttime', 'stoptime', 'start station id', 'start station name', 'start station latitude', 'start station longitude', 'end station id', 'end station name', 'end station latitude', 'end station longitude', 'bikeid', 'usertype', 'birth year', 'gender']
{'tripduration': '4225', 'starttime': '2020-09-01 00:00:01.0430', 'stoptime': '2020-09-01 01:10:26.6350', 'start station id': '3508', 'start station name': 'St Nicholas Ave & Manhattan Ave', 'start station latitude': '40.809725', 'start station longitude': '-73.953149', 'end station id': '116', 'end station name': 'W 17 St & 8 Ave', 'end station latitude': '40.74177603', 'end station longitude': '-74.00149746', 'bikeid': '44317', 'usertype': 'Customer', 'birth year': '1979', 'gender': '1'}
{'tripduration': '1868', 'starttime': '2020-09-01 00:00:04.8320', 'stoptime': '2020-09-01 00:31:13.7650', 'start station id': '3621', 'start station name': '27 Ave & 9 St', 'start station latitude': '40.7739825', 'start station longitude': '-73.9309134', 'end station id': '3094', 'end station name': 'Graham Ave & Withers St', 'end station latitude': '40.7169811', 'end station longitude': '-73.94485918', 'bikeid': '37793', 'usertype': 'Customer', 'birth year': '1991', 'gender': '1'}
{'tripduration': '1097', 'starttime': '2020-09-01 00:00:06.8990', 'stoptime': '2020-09-01 00:18:24.2260', 'start station id': '3492', 'start station name': 'E 118 St & Park Ave', 'start station latitude': '40.8005385', 'start station longitude': '-73.9419949', 'end station id': '3959', 'end station name': 'Edgecombe Ave & W 145 St', 'end station latitude': '40.823498', 'end station longitude': '-73.94386', 'bikeid': '41438', 'usertype': 'Subscriber', 'birth year': '1984', 'gender': '1'}
{'tripduration': '1473', 'starttime': '2020-09-01 00:00:07.7440', 'stoptime': '2020-09-01 00:24:41.1800', 'start station id': '3946', 'start station name': 'St Nicholas Ave & W 137 St', 'start station latitude': '40.818477', 'start station longitude': '-73.947568', 'end station id': '4002', 'end station name': 'W 144 St & Adam Clayton Powell Blvd', 'end station latitude': '40.820877', 'end station longitude': '-73.939249', 'bikeid': '35860', 'usertype': 'Customer', 'birth year': '1990', 'gender': '2'}
{'tripduration': '1193', 'starttime': '2020-09-01 00:00:12.2020', 'stoptime': '2020-09-01 00:20:05.5470', 'start station id': '3081', 'start station name': 'Graham Ave & Grand St', 'start station latitude': '40.711863', 'start station longitude': '-73.944024', 'end station id': '3048', 'end station name': 'Putnam Ave & Nostrand Ave', 'end station latitude': '40.68402', 'end station longitude': '-73.94977', 'bikeid': '26396', 'usertype': 'Customer', 'birth year': '1969', 'gender': '0'}

```

2. Example 4-3:

Input 4-3

```

main.py > ...
1  #=====
2  #Title: Week 4.3 Exercise
3  #Author: Susan E. McGregor
4  #Date: 29 June 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates the use
7  #a simple example of how to read data from a .txt source file
8  #using Python's built-in `csv` library.
9  #=====
10
11 #Import the `csv` library
12 import csv
13
14 #Open the source_file and store it in a variable called `txt_source_file`
15 txt_source_file = open("ShugermanProsecutorPoliticians-
    SupremeCourtJustices.txt","r")
16
17 #Create a `csv.DictReader` object and store it in a variable called
18 #`politicians_reader` with a tab character delimiter of `\t`
19 politicians_reader = csv.DictReader(txt_source_file, delimiter='\t')
20
21 #Print the fieldnames of the `politicians_reader` object in the header row
22 print(politicians_reader.fieldnames)
23
24 #Print the first row of the `politicians_reader` object
25 #using the `next()` function.
26 print(next(politicians_reader))

```

Output 4-3

```

['', 'Justice', 'Term Start/End', 'Party', 'State', 'Pres Appt', 'Other Offices Held', 'Relevant Prosecutorial
Background']
{'': '40', 'Justice': 'William Strong', 'Term Start/End': '1870-1880', 'Party': 'D/R', 'State': 'PA', 'Pres Ap
pt': 'Grant', 'Other Offices Held': 'US House, Supr Court of PA, elect comm for elec of 1876', 'Relevant Prose
cutorial Background': 'lawyer'}

```

3. Example 4-4:

Input 4-4

```
main.py > ...
1  #=====
2  #Title: Week 4.4 Exercise
3  #Author: Susan E. McGregor
4  #Date: 29 June 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates an example of reading
7  #data from an .xlsx file with Python, using the "openpyxl" library
8  #and writing the data to a .csv file
9  #=====
10
11 #Import the 'openpyxl' library, to read our .xlsx file
12 #and the 'load_workbook()' recipe from it
13 from openpyxl import load_workbook
14
15 #Import the 'csv' library, to write our .csv output file
16 import csv
17
18 #Load the .xlsx file into a variable called 'source_workbook'
19 #Note that we're using the 'load_workbook()' recipe from the 'openpyxl' library
20 source_workbook = load_workbook(filename = 'fredgraph.xlsx')
21
22 #Print the names of the worksheets in 'source_workbook' for reference
23 print(source_workbook.sheetnames)
24
25 #Loop through the names of the worksheets in 'source_workbook'
26 for sheet_num, sheet_name in enumerate(source_workbook.sheetnames):
27
28     #Create a variable called 'current_sheet' that holds the current worksheet
29     current_sheet = source_workbook[sheet_name]
30
31     #Print the name of the current worksheet for reference
32     print(sheet_name)
33
34     #Create a new .csv file to write our data to
35     output_file = open("xlsx_"+sheet_name+".csv", "w")
36
```

```

37     #Use the `csv.writer()` recipe from the `csv` library to create
38     #a writer object to write to our output file
39     output_writer = csv.writer(output_file)
40
41     #Loop through the rows in the current worksheet
42     for row in current_sheet.iter_rows():
43
44         #Create a new list to hold the values in the current row
45         row_cells = []
46
47         #Loop through the cells in the current row
48         for cell in row:
49
50             #Print the cell and its value for reference
51             print(cell, cell.value)
52
53             #Add the value of the current cell to our `row_cells` list
54             row_cells.append(cell.value)
55
56         #Write the `row_cells` list to our output file, as a new row
57         output_writer.writerow(row_cells)
58
59     #Close the output file
60     output_file.close()
61

```

Output Console 4-4

```

ell 'FRED Graph'.A690> 2004-06-01 00:00:00
<Cell 'FRED Graph'.B690> 9.6
<Cell 'FRED Graph'.C690> 5.6
<Cell 'FRED Graph'.A691> 2004-07-01 00:00:00
<Cell 'FRED Graph'.B691> 9.5
<Cell 'FRED Graph'.C691> 5.5
<Cell 'FRED Graph'.A692> 2004-08-01 00:00:00
<Cell 'FRED Graph'.B692> 9.4
<Cell 'FRED Graph'.C692> 5.4
<Cell 'FRED Graph'.A693> 2004-09-01 00:00:00
<Cell 'FRED Graph'.B693> 9.4
<Cell 'FRED Graph'.C693> 5.4
<Cell 'FRED Graph'.A694> 2004-10-01 00:00:00
<Cell 'FRED Graph'.B694> 9.7
<Cell 'FRED Graph'.C694> 5.5
<Cell 'FRED Graph'.A695> 2004-11-01 00:00:00
<Cell 'FRED Graph'.B695> 9.4
<Cell 'FRED Graph'.C695> 5.4
<Cell 'FRED Graph'.A696> 2004-12-01 00:00:00
<Cell 'FRED Graph'.B696> 9.3

```

Output File 4-4

xlsx_FRED Graph.csv			
Search			
	FRED Graph Observations		
🔗	Federal Reserve Economic Data		
🔗	Link: https://fred.stlouisfed.org		
🔗	Help: https://fredhelp.stlouisfed.org		
🔗	Economic Research Division		
🔗	Federal Reserve Bank of St. Louis		
🔗			
🔗	U6RATE	Total Unemployed, Plus All Persons Marginally Attache	
🔗	UNRATE	Unemployment Rate, Percent, Monthly, Seasonally Adj	
🔗			
🔗	Frequency: Monthly		
🔗	observation_date	U6RATE	UNRATE
🔗	1948-01-01 00:00:00		3.4
🔗	1948-02-01 00:00:00		3.8
🔗	1948-03-01 00:00:00		4.0
🔗	1948-04-01 00:00:00		3.9
🔗	1948-05-01 00:00:00		3.5
🔗	1948-06-01 00:00:00		3.6
🔗	1948-07-01 00:00:00		3.6
🔗	1948-08-01 00:00:00		3.9
🔗	1948-09-01 00:00:00		3.8
🔗	1948-10-01 00:00:00		3.7
🔗	1948-11-01 00:00:00		3.8
🔗	1948-12-01 00:00:00		4.0
🔗	1949-01-01 00:00:00		4.3
🔗	1949-02-01 00:00:00		4.7
🔗	1949-03-01 00:00:00		5.0
🔗	1949-04-01 00:00:00		5.3
🔗	1949-05-01 00:00:00		6.1
🔗	1949-06-01 00:00:00		6.2
🔗	1949-07-01 00:00:00		6.5

4. Example 4-6

Input 4-6

```

main.py > ...
1  #=====
2  #Title: Week 4.6 Exercise
3  #Author: Susan E. McGregor
4  #Date: 29 June 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates an example of reading
7  #data from a .xls file with Python using the "xlrd" library.
8  #=====
9
10 #Import the "xlrd" library
11 import xlrd
12
13 #Import the "csv" library for writing to a .csv file
14 import csv
15
16 #Pass the "fredgraph.xls" file to the 'open_workbook' recipe
17 #to create a variable that points to the file
18 source_workbook = xlrd.open_workbook("fredgraph.xls")
19
20 #Loop through every sheet in the workbook
21 for sheet_name in source_workbook.sheet_names():
22
23     #Create a variable that points to the current sheet by
24     #using the 'sheet_by_name' recipe
25     current_sheet = source_workbook.sheet_by_name(sheet_name)
26
27     #Print the name of the current sheet to the console
28     print(sheet_name)
29
30     #Create a new '.csv' file with the same name as the current sheet
31     #and open it for writing
32     output_file = open("xls_"+sheet_name+".csv","w")
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
34     #Use the 'csv.writer' recipe to create a writer object
35     #that will write to the output file
36     output_writer = csv.writer(output_file)
37
38     #Loop through every row in the current sheet
39     #the 'get_rows' recipe returns a list of rows
40     #each of which is a list of cells
41     for row_num, row in enumerate(current_sheet.get_rows()):
42
43         #Write the current row to the output file
44         #the 'row_values' recipe returns a list of values
45         #for the current row
46         #the 'writerow' recipe writes a list of values to the output file
47         output_writer.writerow(current_sheet.row_values(row_num))
48
49     #Close the output file
50     output_file.close()
51

```

Output Console 4-6

FRED Graph

Output File 4-6

xls_FRED Graph.csv		
Search		
	FRED Graph Observations	
	Federal Reserve Economic Data	
	Link: https://fred.stlouisfed.org	
	Help: https://fredhelp.stlouisfed.org	
	Economic Research Division	
	Federal Reserve Bank of St. Louis	
		
	U6RATE	Total Unemployed, Plus All Persons Marginally Attache
		
	Frequency: Monthly	
	observation_date	U6RATE
	34335.0	11.7
	34366.0	11.4
	34394.0	11.5
	34425.0	11.3
	34455.0	10.9
	34486.0	11.0
	34516.0	10.8
	34547.0	10.6
	34578.0	10.4
	34608.0	10.4
	34639.0	10.1
	34669.0	10.1
	34700.0	10.1
	34731.0	9.9
	34759.0	9.9

5. Example 4-7:

Input 4-7

```
main.py > ...
1  #=====
2  #Title: Week 4.7 Exercise
3  #Author: Susan E. McGregor
4  #Date: 29 June 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates an example of reading
7  #data from a fixed-width file with Python and writing it to a
8  #CSV file.
9  #=====
10
11 #Import the csv library to use the csv file writing functionality
12 import csv
13
14 #The name of the file we want to read from
15 filename = "ghcnd-stations"
16
17 #Open the file for reading and store a reference to the file in a variable
18 #called source_file
19 source_file = open(filename+".txt", "r")
20
21 #Read all of the lines from the file and store them in a list
22 stations_list = source_file.readlines()
23
24 #Create a output file to write the data to
25 output_file = open(filename+".csv","w")
26
27 #Use the csv library to create a writer object for writing data to the output file
28 output_writer = csv.writer(output_file)
29
30 #Create the header list
31 headers = ["ID","LATITUDE","LONGITUDE","ELEVATION","STATE","NAME","GSN_FLAG",
32            "HCNCRN_FLAG","WMO_ID"]
33
34 #Use the writer object to write the header list to the output file
35 output_writer.writerow(headers)
```

```
36
37 #Loop through each line in the list of lines
38 for line in stations_list:
39     #Create a new list to store the data for the current row
40     new_row = []
41     #ID: positions 1-11
42     new_row.append(line[0:11])
43     #LATITUDE: positions 13-20
44     new_row.append(line[12:20])
45     #LONGITUDE: positions 22-30
46     new_row.append(line[21:30])
47     #ELEVATION: positions 32-37
48     new_row.append(line[31:37])
49     #STATE: positions 39-40
50     new_row.append(line[38:40])
51     #NAME: positions 42-71
52     new_row.append(line[41:71])
53     #GSN_FLAG: positions 73-75
54     new_row.append(line[72:75])
55     #HCNCRN_FLAG: positions 77-79
56     new_row.append(line[76:79])
57     #WMO_ID: positions 81-85
58     new_row.append(line[80:85])
59
60     #Use the writer object to write the new row to the output file
61     output_writer.writerow(new_row)
62
63 #Close the output file to save the data
64 output_file.close()
```

Output File 4-7

ghcnd-stations.csv

Search

ID	LATITUDE	LONGITUDE	ELEVATION	STATE	NAME	GSN_FLAG	HCNCRN_FLAG	WMO_ID
ACW00011604	17.1167	-61.7833	10.1		ST JOHNS COOLIDGE FLD			
ACW00011647	17.1333	-61.7833	19.2		ST JOHNS			
AE000041196	25.3330	55.5170	34.0		SHARJAH INTER. AIRP	GSN		41196
AEM00041194	25.2550	55.3640	10.4		DUBAI INTL			41194
AEM00041217	24.4330	54.6510	26.8		ABU DHABI INTL			41217
AEM00041218	24.2620	55.6090	264.9		AL AIN INTL			41218
AF000040930	35.3170	69.0170	3366.0		NORTH-SALANG	GSN		40930
AFM00040938	34.2100	62.2280	977.2		HERAT			40938
AFM00040948	34.5660	69.2120	1791.3		KABUL INTL			40948
AFM00040990	31.5000	65.8500	1010.0		KANDAHAR AIRPORT			40990
AG000060390	36.7167	3.2500	24.0		ALGER-DAR EL BEIDA	GSN		60390
AG000060590	30.5667	2.8667	397.0		EL-GOLEA	GSN		60590
AG000060611	28.0500	9.6331	561.0		IN-AMENAS	GSN		60611
AG000060680	22.8000	5.4331	1362.0		TAMANRASSET	GSN		60680
AGE00135039	35.7297	0.6500	50.0		ORAN-HOPITAL MILITAIRE			
AGE00147704	36.9700	7.7900	161.0		ANNABA-CAP DE GARDE			
AGE00147705	36.7800	3.0700	59.0		ALGIERS-VILLE/UNIVERSITE			
AGE00147706	36.8000	3.0300	344.0		ALGIERS-BOUZAREAH			
AGE00147707	36.8000	3.0400	38.0		ALGIERS-CAP CAXINE			
AGE00147708	36.7200	4.0500	222.0		TIZI OUZOU			60395
AGE00147709	36.6300	4.2000	942.0		FORT NATIONAL			
AGE00147710	36.7500	5.1000	9.0		BEJAIA-BOUGIE (PORT)			60401
AGE00147711	36.3697	6.6200	660.0		CONSTANTINE			
AGE00147712	36.1700	1.3400	112.0		ORLEANSVILLE (CHLEF)			

ghcnd-stations - Notepad

File Edit Format View Help

ID	LATITUDE	LONGITUDE	ELEVATION	STATE	NAME	GSN_FLAG	HCNCRN_FLAG	WMO_ID
ACW00011604	17.1167	-61.7833	10.1		ST JOHNS COOLIDGE FLD			
ACW00011647	17.1333	-61.7833	19.2		ST JOHNS			
AE000041196	25.3330	55.5170	34.0		SHARJAH INTER. AIRP	GSN		41196
AEM00041194	25.2550	55.3640	10.4		DUBAI INTL			41194
AEM00041217	24.4330	54.6510	26.8		ABU DHABI INTL			41217
AEM00041218	24.2620	55.6090	264.9		AL AIN INTL			41218
AF000040930	35.3170	69.0170	3366.0		NORTH-SALANG	GSN		40930
AFM00040938	34.2100	62.2280	977.2		HERAT			40938
AFM00040948	34.5660	69.2120	1791.3		KABUL INTL			40948
AFM00040990	31.5000	65.8500	1010.0		KANDAHAR AIRPORT			40990
AG000060390	36.7167	3.2500	24.0		ALGER-DAR EL BEIDA	GSN		60390
AG000060590	30.5667	2.8667	397.0		EL-GOLEA	GSN		60590
AG000060611	28.0500	9.6331	561.0		IN-AMENAS	GSN		60611
AG000060680	22.8000	5.4331	1362.0		TAMANRASSET	GSN		60680
AGE00135039	35.7297	0.6500	50.0		ORAN-HOPITAL MILITAIRE			
AGE00147704	36.9700	7.7900	161.0		ANNABA-CAP DE GARDE			
AGE00147705	36.7800	3.0700	59.0		ALGIERS-VILLE/UNIVERSITE			
AGE00147706	36.8000	3.0300	344.0		ALGIERS-BOUZAREAH			
AGE00147707	36.8000	3.0400	38.0		ALGIERS-CAP CAXINE			
AGE00147708	36.7200	4.0500	222.0		TIZI OUZOU			60395
AGE00147709	36.6300	4.2000	942.0		FORT NATIONAL			
AGE00147710	36.7500	5.1000	9.0		BEJAIA-BOUGIE (PORT)			60401
AGE00147711	36.3697	6.6200	660.0		CONSTANTINE			
AGE00147712	36.1700	1.3400	112.0		ORLEANSVILLE (CHLEF)			
AGE00147713	36.1800	5.4000	1081.0		SETIF			
AGE00147714	35.7700	0.8000	78.0		ORAN-CAP FALCON			
AGE00147715	35.4200	8.1197	863.0		TEBESSA			
AGE00147716	35.1000	-1.8500	83.0		NEMOURS (GHAZAQUET)			60517

Ln 17, Col 86 100% Windows (CRLF)

6. Example 4-15:

Input 4-15

```
main.py x json_U6_FRED_data.csv +
main.py > ...

1  #=====
2  #Title: Week 4.15 Exercise
3  #Author: Susan E. McGregor
4  #Date: 29 June 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates an example of reading
7  #data from a .json file with Python, using the built-in "json"
8  #library. The data used here is an instance of
9  #https://api.stlouisfed.org/fred/series/observations?series_id=U6RATE& \
10 #file_type=json&api_key=YOUR_API_KEY_HERE
11 #=====
12
13 #Import the json library to read the .json file
14 import json
15
16 #Import the csv library to write the .csv file
17 import csv
18
19 #Make a variable to hold the filename
20 filename = "U6_FRED_data"
21
22 #Open the .json file in read mode and assign it to a variable
23 json_source_file = open(filename+".json","r")
24
25 #Use the json library's load function to read the data
26 #from the .json file into a variable called json_data
27 json_data = json.load(json_source_file)
28
29 #Create a new .csv file in write mode and assign it to a variable
30 output_file = open("json_"+filename+".csv","w")
31
32 #Use the csv library's writer function to create a writer object
33 #that will write to the .csv file we just created
34 output_writer = csv.writer(output_file)
35
```

```

36 #Use the writer object's writerow function to write the header row
37 output_writer.writerow(list(json_data["observations"][0].keys()))
38
39 #Loop through each object in the json_data["observations"] list
40 for obj in json_data["observations"]:
41
42     #Create an empty list to hold the values of each object
43     obj_values = []
44
45     #Loop through each key, value pair in the object
46     for key, value in obj.items():
47
48         #Print the key and value to the console
49         print(key,value)
50
51         #Append the value to the obj_values list
52         obj_values.append(value)
53
54     #Use the writer object's writerow function to write
55     #the obj_values list to the .csv file
56     output_writer.writerow(obj_values)
57
58 #Close the .csv file to save it
59 output_file.close()

```

Output Console 4-15

```

realtime_end 2020-12-30
date 1999-08-01
value 7.3
realtime_start 2020-12-30
realtime_end 2020-12-30
date 1999-09-01
value 7.4
realtime_start 2020-12-30
realtime_end 2020-12-30
date 1999-10-01
value 7.2
realtime_start 2020-12-30
realtime_end 2020-12-30
date 1999-11-01
value 7.2
realtime_start 2020-12-30
realtime_end 2020-12-30
date 1999-12-01
value 7.1
realtime_start 2020-12-30

```

Output File 4-15

json_U6_FRED_data.csv

Search

	realtime_start	realtime_end	date	value
/	2020-12-30	2020-12-30	1994-01-01	11.7
/	2020-12-30	2020-12-30	1994-02-01	11.4
/	2020-12-30	2020-12-30	1994-03-01	11.5
/	2020-12-30	2020-12-30	1994-04-01	11.3
/	2020-12-30	2020-12-30	1994-05-01	10.9
/	2020-12-30	2020-12-30	1994-06-01	11.0
/	2020-12-30	2020-12-30	1994-07-01	10.8
/	2020-12-30	2020-12-30	1994-08-01	10.6
/	2020-12-30	2020-12-30	1994-09-01	10.4
/	2020-12-30	2020-12-30	1994-10-01	10.4
/	2020-12-30	2020-12-30	1994-11-01	10.1
/	2020-12-30	2020-12-30	1994-12-01	10.1
/	2020-12-30	2020-12-30	1995-01-01	10.1
/	2020-12-30	2020-12-30	1995-02-01	9.9
/	2020-12-30	2020-12-30	1995-03-01	9.9
/	2020-12-30	2020-12-30	1995-04-01	10.1
/	2020-12-30	2020-12-30	1995-05-01	10.1
/	2020-12-30	2020-12-30	1995-06-01	10.1
/	2020-12-30	2020-12-30	1995-07-01	10.1
/	2020-12-30	2020-12-30	1995-08-01	10.1
/	2020-12-30	2020-12-30	1995-09-01	10.2
/	2020-12-30	2020-12-30	1995-10-01	10.0
/	2020-12-30	2020-12-30	1995-11-01	10.1
/	2020-12-30	2020-12-30	1995-12-01	10.0
/	2020-12-30	2020-12-30	1996-01-01	9.8
/	2020-12-30	2020-12-30	1996-02-01	10.0

7. Example 4-16

Input 4-16

```

main.py > ...
1  #=====
2  #Title: Week 4.16 Exercise
3  #Author: Susan E. McGregor
4  #Date: 29 June 2025
5  #Modified by: Wade Eckert
6  #Description: This Program demonstrates an example of reading
7  #data from a .pdf file with Python, using 'pdf2image' to convert
8  #it to images, and then using the openCV and 'tesseract' libraries
9  #to extract the text from the images.
10 # The source data was downloaded from:
11 # https://files.stlouisfed.org/files/htdocs/publications/page1-econ/2020/12/01/ \
12 # unemployment-insurance-a-tried-and-true-safety-net_SE.pdf
13 #=====
14
15 #Use the 'os' library to create a new folder to store our images and text
16 import os
17
18 #Use the 'pdf2image' library to convert our pdf to images
19 from pdf2image import convert_from_path
20
21 #Use the 'glob' library to find all the files in our target
22 #folder with a certain file extension and loop through them
23 import glob
24
25 #Use the 'cv2' library to interpret our images
26 import cv2
27
28 #Use the 'pytesseract' library to extract text from our images
29 import pytesseract
30
31 #Set the name of the pdf we want to convert to images
32 pdf_name = "SafetyNet"
33
34 #Create a variable to hold the name of the pdf we want to convert
35 pdf_source_file = pdf_name+".pdf"
36
37 #Creat if statement for ass long as a folder with the same name
38 #as the pdf does not already exist
39 if os.path.isdir(pdf_name) == False:
40     #Create a new folder with the same name as the pdf
41     target_folder = os.mkdir(pdf_name)
42
43 #Create a variable to hold the converted images from our pdf
44 pages = convert_from_path(pdf_source_file, 300)
45
46 #Loop through all the pages in our pdf and save each one as an image
47 for page_num, page in enumerate(pages):
48     #Create unique filenames for each page image, combining the
49     #Folder name and the page number
50     filename = os.path.join(pdf_name,"p"+str(page_num)+".png")
51     #Save each page as an image in our target folder
52     page.save(filename, 'PNG')
53

```

```

54 #Loop through all the images in our target folder that have a '.png' file extension
55 #and extract the text from them using the 'cv2' and 'pytesseract' libraries
56 #and save the extracted text to a new file in our target folder with the same name
57 for img_file in glob.glob(os.path.join(pdf_name, '*.png')):
58     #Replace the slash in the image's filename with a dot
59     temp_name = img_file.replace("/", ".")
60     #Pull the unique page name (e.g. 'p2') from the 'temp_name'
61     #variable by splitting the string at the dot and taking the second element
62     text_filename = temp_name.split(".")[1]
63     #Create a new file to hold the extracted text from the image
64     output_file = open(os.path.join(pdf_name, text_filename + ".txt"), "w")
65     #Use the 'cv2' library to read the image file
66     img = cv2.imread(img_file)
67     #Create a variable to hold the extracted text from the image
68     converted_text = pytesseract.image_to_string(img)
69     #Write the extracted text to the new file
70     output_file.write(converted_text)
71     #Close the output file to save it
72     output_file.close()

```

Output Directory 14.6

data_wrangling > data_wrangling_exercises > Chapter_4 > SafetyNet			
	Name	Status	Date n
al	 p0		6/29/2
	 p0		6/29/2
	 p1		6/29/2
	 p1		6/29/2
	 p2		6/29/2
	 p2		6/29/2
	 p3		6/29/2
	 p3		6/29/2
	 p4		6/29/2
	 p4		6/29/2
	 p5		6/29/2
	 p5		6/29/2
	 p6		6/29/2
	 p6		6/29/2
	 p7		6/29/2
	 p7		6/29/2

Output Screenshots 14.6

Focus on **FINANCE**

GLOSSARY

Depression: A period of time lasting several years in which economic activity is depressed and unemployment is high.

Employee: A person who works for an employer in exchange for a monetary payment.

Employer: A person or business providing a job to a worker in exchange for money.

Extended unemployment benefits: A type of unemployment benefit that provides additional income to workers during periods of high unemployment.

Labor force: The total number of people who are able to work and are currently working or looking for work.

Massive loss of income: A period of time in which many people lose their jobs and income, leading to widespread poverty and hardship.

Taxes: Fees charged on businesses and individuals to fund government services and infrastructure.

Unemployed: People who are out of work and are actively looking for a job.

Unemployment compensation: A type of benefit that provides income to workers who have lost their jobs through no fault of their own.

Unemployment rate: The percentage of the labor force that is unemployed.

PAGE ONE Economics™

**Unemployment Insurance:
A Tried and True Safety Net**

Jeanette N. Bennett, Senior Economic Educator, Spokesman

"Of all the aspects of social policy making, no one is looking at unemployment."

What happens when workers lose their jobs through no fault of their own? How do they manage with the loss of income? Today, every state has an unemployment insurance (UI) program that provides some income to these **unemployed workers**. But how did it begin and how does it work?

The prosperous times of the 1920s set an expectation of having employment and money to spend. Electricity, radios, refrigerators, and cars became more affordable. And, with employment opportunities, people grew accustomed to the good times of the "Roaring Twenties" and thought they would last forever.

But beginning in 1929, things changed. The Great Depression brought an economic catastrophe with a challenging crisis. Between 1929 and 1933, the **unemployment rate** soared to 25 percent of the **labor force** (Figure 1). The effects were alarming and in stark contrast to the 1920s. Without a safety net, a lost job meant no income. Many became homeless with no means to support their families. And the struggles began.

**Figure 1
Civilian Unemployment Rate, 1912-2005**

SOURCE: Federal Reserve Bank of St. Louis.
<https://www.federalreserve.gov/data/fred/series/unemployment-rate>

December 2020

Federal Reserve Bank of St. Louis | www.federalreserve.org

```

SafetyNet > p0.txt

1  Focus on FINANCE
2
3  GLOSSARY
4
5  Depression: A severe and long-lasting eco-
6  nomic downturn that is worse and deeper
7  than a recession; a severe reduction in
8  gross domestic product (GDP).
9
10 Employee: A person who works for an
11 employer in exchange for a monetary
12 payment.
13
14 Employer: A person or business providing a
15 job or work to others and giving a mone-

```

PAGE ONE Economics

Federal Reserve Bank of St. Louis | Research and Education | 2

Federal-State Partnership

Struggles and crises can bring innovative ideas and reforms. And that's exactly what happened during the Great Depression. The ugly picture of severe **depression** with despair and poverty brought a renewed interest in and attention to providing **unemployment compensation** (UC).

Programs were tried at the state level. Wisconsin led the way and in 1932 became the first state to enact a UI law. A few other states followed with similar programs, which were funded by a **tax on employers**.¹ But most states did not. They feared a loss of business and jobs to states that did not tax employers. This created an interstate problem and was, therefore, a direct concern of the federal government.

In 1935, as part of the Social Security Act, the Federal Unemployment Tax Act (FUTA) created the federal-state unemployment insurance program.² This act mirrored the earlier state plans with the purpose of temporarily replacing a portion of wages for workers who had been laid off and who were looking and available for work.

The act gave oversight responsibility to the U.S. Department of Labor (DOL). With federal guidelines, states were given the freedom and flexibility to set criteria and design their programs. State unemployment offices were created to implement their UI programs and make payments to workers who qualified.

Today, all states have UI programs, but they vary in design. For example, eligibility requirements for receiving benefits vary based on duration of prior employment and whether full-time or part-time status. Also, the normal maximum amount of benefits, the duration of benefits and claim procedures often differ. And UI program guidelines within each state are primarily subject to change according to economic conditions; this can be done by state governments or by the federal government in times of recession or special conditions. (See boxed insert, "Sample of Normal State Unemployment Benefits, August 2020.")³

Eligibility for Unemployment Insurance

All states require that workers be able to work, available for work, and actively seeking work for UI eligibility. Most states require a work history listing a minimum time

Sample of Normal State Unemployment Benefits, August 2020

State	Normal maximum weeks of benefits	Normal maximum weekly benefits
Arkansas	26	\$451
California	26	\$410
Florida	7	\$273
Georgia	12	\$455
Indiana	26	\$290
Kentucky	16	\$311
Louisiana	26	\$177
Massachusetts	30	\$353
Mississippi	26	\$455
Missouri	20	\$120
Nebraska	16	\$409
North Dakota	26	\$470
Oklahoma	26	\$129
Tennessee	16	\$475
Texas	26	\$121
Washington	26	\$564

workers and a minimum amount of earnings based on a 12-month period.⁴

Workers must file claims weekly to maintain eligibility. Any earnings for the period, including job offers accepted or declined, must be reported. This can affect eligibility for receiving benefits and the amount received. Additionally, workers must continue to meet state requirements.⁵

In all states, workers can be denied benefits. Knowingly making false statements to obtain benefit payments is one reason for denial. Other reasons include voluntarily leaving work without good cause and being disqualified for misconduct connected with work. And even after being determined eligible for UI, workers can become ineligible if they're unable to or unavailable for work, not actively seeking work or refusing an offer of suitable work.⁶ It does not mean, though, to leave their jobs voluntarily or people looking for their first job. Individuals who are self-employed are normally not entitled to regular UC. Also, certain types of employment, such as charitable work, are excluded from UI coverage.⁷

Benefits

Each state has its own formula for determining unemployment benefits. And the number of unemployed workers

```
SafetyNet > p1.txt

1  PAGE ONE Economics®
2
3  Federal-State Partnership
4
5  Struggles and crises can bring innovative ideas and
6  reform. And that's exactly what happened during the
7  Great Depression. The ugly picture of severe depression
8  with despair and poverty brought a renewed interest
9  in and attention to providing unemployment
10 compensation (UQ).
11
12 Programs were tried at the state level. Wisconsin led the
13 way and in 1932 became the first state to enact a UI law.
14 A few other states followed with similar programs, which
15 were funded by a tax on employers. But most states
16 did not. They feared a loss of business and jobs to states
```

The regular UI program is generally funded by employers. First, employers pay both federal and state unemployment taxes.¹² The unemployment taxes are based on the amount of wages paid to employees and are determined as a percentage of an employee's wages. The "UI" tax

1 PAGE ONE Economics®

2

3 Figure 2

4

5 Federal Reserve Bank of St. Louis | research.stlouisfed.org 3

6

7 Percent of Labor Force Getting Unemployment Benefits, August 27, 2020

8

9 Percent of labor force
10 getting benefits

11

12 | =15%

13 [T 11-14%

14 [] 6-10%

15 [] =5%

16

17 SOURCE: Stateline analysis of U.S. Department of Labor and U.S. Bureau of Economic Analysis

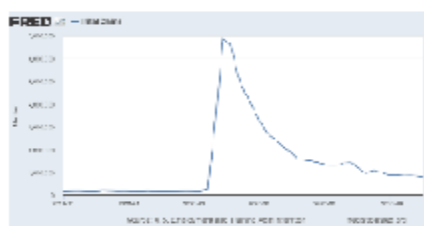
18

19 receiving UI changes according to economic conditions

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Figure 3
Initial Claims



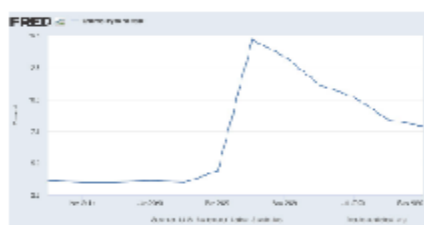
SC130101000000 Federal Reserve Bank of St. Louis
<https://fred.stlouisfed.org/series/SC130101000000>

Figure 4
Continued Claims (Insured Unemployment)



SC130101000000 Federal Reserve Bank of St. Louis
<https://fred.stlouisfed.org/series/SC130101000000>

Figure 5
Unemployment Rate



SC130101000000 Federal Reserve Bank of St. Louis
<https://fred.stlouisfed.org/series/SC130101000000>

rate is the same for all employers in all states, however, the state employment tax varies from one state to another.

Employers must pay FUTA tax on a regular basis. And at the end of a year, employers must file a FUTA tax return. This tax return requires filers to record FUTA tax payments made for the year and submit any payment due. FUTA taxes are submitted to the Internal Revenue Service (IRS) and are credited to individual state accounts.

As part of the federal-state partnership, the federal government pays for administrative costs and for setting up employment offices that attempt to match workers with new jobs.¹⁸ Determining benefits and funding payments are state responsibilities.

Data Collection

Data collected by the DOL from state unemployment agencies for the number of UI claims filed weekly are helpful. These data can tell something about the job market. For example, when the number of initial claims is steep, it's an indication of a abrupt loss of jobs (Figure 3). Continued claims (Figure 4) indicate the number of unemployed people who have already filed an initial claim and are filing a claim so that they keep receiving weekly benefits.

However, many workers who lose their job may not file for UI for a while. And many people remain unemployed after benefits expire. So, UI claims data do not precisely show the number of unemployed persons and are not a measure of the unemployment rate. The unemployment rate is measured quite differently by another agency, the Bureau of Labor Statistics (BLS). The BLS does not use the unemployment claims data to figure the unemployment rate. It uses a system of household surveys to collect information from a representative sample of the U.S. population to calculate the unemployment rate (Figure 5).

Unemployment Insurance Changes

UI programs are subject to ongoing changes. For example, over the years the duration of regular UI benefits has increased from 16 weeks to 26 weeks in most states. Also, in some previous UI programs, workers could become eligible for UI after being fired from their job or quitting work. Over the years this became disallowed in all states.¹⁹ Technology has also changed the UI process. In earlier

SafetyNet > p3.txt

```

1 PAGE ONE Economics®
2
3 Figure 3
4 Initial Claims
5
6 FRED ~/- Initial Claims
7 7,000,000
8 6,000,000
9 5,000,000
10
11 4,000,000
12
13 Number
14
15 3,000,000
16 2,000,000

```


Name _____ Period _____

Federal Reserve Bank of St. Louis Page One Economics®

"Unemployment Insurance: A Tried and True Safety Net"

After reading the article, select the best answer to each question.

- The first unemployment insurance program
 - began at the federal level.
 - was funded by a tax on employees.
 - was designed by the Internal Revenue Service.
 - began at the state level.
- Federal unemployment taxes are sent to
 - the Bureau of Labor Statistics.
 - the Internal Revenue Service.
 - the Department of Labor.
 - state unemployment offices.
- Data collected by the BLS for weekly UI claims
 - come from the Bureau of Labor Statistics.
 - are based on data from state agencies.
 - will always show the unemployment rate.
 - are the same for each week.
- UI is generally funded by
 - taxes on employees.
 - taxes on employers.
 - employees and unemployed workers.
 - a policy that employees can buy.
- Although amounts often change, the normal maximum weekly benefits a worker could receive from UI in August 2000 was
 - the same in Tennessee and Washington.
 - the same in Tennessee and Florida.
 - more in Mississippi than in Texas.
 - the same in every state.
- To qualify for UI benefits, a worker
 - can refuse to work.
 - can be sick and in the hospital.
 - must be able and available to work.
 - must file for benefits each day.

SafetyNet > p5.txt

```

1  PAGE ONE Economics® Federal Reserve Bank of St. Louis | research.stlouisfed.org 6
2
3  Name Period
4
5  Federal Reserve Bank of St. Louis Page One Economics®:
6  "Unemployment Insurance: A Tried and True Safety Net"
7
8  After reading the article, select the best answer to each question.
9
10 1. The first unemployment insurance program
11   a. began at the federal level.
12   b. was funded by a tax on employees.
13   c. was designed by the Internal Revenue Service.
14   d. began at the state level.
15
16 2. Federal unemployment taxes are sent to

```

7. An unemployed worker received a weekly benefit of \$500 in early March 2020. If the worker remained eligible, how much was the weekly benefit at the beginning of April 2020?
- \$1,100
 - \$750
 - \$400
 - \$600
8. Data collected by the DOL from state agencies for both initial and continued unemployment insurance claims
- are helpful in learning about the job market.
 - show exactly the same things.
 - are used for arriving at the unemployment rate.
 - remain unchanged from week to week.
9. Federal unemployment taxes are called "FUTA" taxes and are
- determined by the state in which a worker works.
 - paid only by an employee to the IRS.
 - paid by employers to the DOL.
 - based on the amount of wages paid to employees.
10. The unemployment rate increased sharply
- during the Great Depression and decreased when COVID-19 hit in 2020.
 - during the Great Depression and when COVID-19 hit in 2020.
 - during the 1940s and 1950s.
 - during the 1970s and 1980s.
11. Extended unemployment benefits
- can only be given by individual states.
 - can be given by both individual states and the federal government.
 - are only available with federal approval.
 - were not given when COVID-19 hit in 2020.
12. The FUTA was passed in 1935 and gave
- full control of all UI programs to the DOL.
 - full control of the design of individual state UI programs to the BLS.
 - all states the freedom to design their own UI programs.
 - full control of the design of individual state UI programs to the IRS.
13. Most states have
- 26 weeks as the normal maximum number of weeks of benefits.
 - more than \$800 for the normal maximum benefits.
 - less than 12 weeks as the normal maximum number of weeks of benefits.
 - more than 26 weeks as the normal maximum number of weeks of benefits.

SafetyNet > p6.txt

```

1  PAGE ONE Economics® Federal Reserve Bank of St. Louis | research.stlouisfed.org
2
3  10.
4
5  11.
6
7  12.
8
9  13.
10
11  An unemployed worker received a weekly benefit of $500 in early March 2020. If the worker remained eligible,
12  how much was the weekly benefit at the beginning of April 2020?
13
14  a. $1,100
15  b. $750

```


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Federal Reserve Bank of St. Louis | Research and Statistics | 4

14. Juan is unemployed and has filed an unemployment claim for the first time. His claim will be counted:
- as part of the initial claim data for the week he filed.
 - as part of the initial claim and continued claims data for the week he filed.
 - in the total number of continued claims for the week.
 - in the data collected by BLS to calculate the unemployment rate.
15. The UI programs in each state:
- all require that claims be filed the day after a worker becomes unemployed.
 - provide a safety net for workers who lose jobs through no fault of their own.
 - are designed at the federal level and are basically all the same.
 - cannot deny benefits to unemployed workers for any reason.
16. The number of unemployment insurance claims is used to determine the unemployment rate.
- True
 - False
17. The number of initial unemployment insurance claims is different from the number of continued claims.
- True
 - False
18. The number of initial unemployment insurance claims decreased sharply in March 2020.
- True
 - False

SafetyNet > p7.txt

```

1 PAGE ONE Economics®
2
3 14. Juan is unemployed and has filed an unemployment claim for the first time. His claim will be counted
4
5 a.
6
7 b.
8 C.
9 d.
10
11 as part of the initial claim data for the week he filed.
12
13 as part of the initial claim and continued claims data for the week he filed.
14 in the total number of continued claims for the week.
15
16 in the data collected by BLS to calculate the unemployment rate.
```